

BADGER, NL, Canada

WMO: 714000

Lat: **48.968N**Lon: **56.061W**Elev: **103**StdP: **100.10**Time Zone: **-3.50 (NAF)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.9	-22.2	-29.6	0.2	-25.4	-25.5	0.4	-21.9	9.5	-4.4	8.5	-3.9	0.7	280	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	28.4	18.7	26.6	18.0	24.8	17.5	21.0	25.5	19.9	23.7	18.8	22.6	3.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	23.1	18.5	13.5	22.1	17.3	12.6	20.8	61.5	25.7	57.5	23.8	53.9	22.5	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.9	6.1	DB	-30.6	32.0	3.0	1.6	-32.8	33.1	-34.5	34.1	-36.2	35.0	-38.4	36.1	(4)
(5)			WB	-30.9	23.0	2.9	0.8	-32.9	23.6	-34.6	24.0	-36.3	24.5	-38.4	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-8.0	-8.4	-4.6	1.3	6.8	11.9	17.0	16.1	11.4	6.0	1.1	-3.9	(6)	
	DBStd	9.80	5.87	6.35	5.74	3.84	3.68	4.19	3.46	3.49	4.06	3.73	4.56	5.46	(7)	
	HDD10.0	2802	558	517	454	261	113	26	1	2	31	137	271	432	(8)	
	HDD18.3	5298	817	750	712	511	356	196	67	86	211	383	518	691	(9)	
	CDD10.0	594	0	0	0	0	16	84	218	191	72	12	2	0	(10)	
	CDD18.3	48	0	0	0	0	0	4	25	17	2	0	0	0	(11)	
	CDH23.3	713	0	0	0	0	17	106	336	219	34	1	0	0	(12)	
	CDH26.7	150	0	0	0	0	3	22	76	41	8	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.7	2.7	2.8	2.8	2.6	2.4	2.2	2.1	2.3	2.3	2.5	2.6	(14)	
(15) Precipitation	PrecAvg	1261	115	105	100	82	85	84	105	109	107	119	119	125	(15)	
	PrecMax	1523	188	171	151	164	131	141	190	174	186	232	198	223	(16)	
	PrecMin	1066	35	40	53	41	31	34	45	46	46	69	64	84	(17)	
	PrecStd	118	35	32	26	30	24	29	36	38	39	45	29	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	8.2	10.4	16.0	25.7	29.2	30.9	30.2	27.2	20.1	15.4	10.6	(19)	
		MCWB	9.2	7.1	6.0	9.5	13.7	17.7	20.0	19.6	18.8	15.6	13.7	9.9	(20)	
	2%	DB	5.6	5.2	6.9	12.4	21.1	26.3	28.7	27.6	23.4	17.6	13.2	7.5	(21)	
		MCWB	4.8	4.2	4.1	7.9	13.0	16.2	19.3	18.8	17.2	14.2	12.0	6.8	(22)	
	5%	DB	2.7	2.6	4.9	10.0	18.1	23.8	26.9	25.8	21.3	15.3	10.9	5.0	(23)	
		MCWB	2.0	1.5	2.8	6.3	11.2	15.5	18.6	18.0	16.7	12.4	9.8	4.3	(24)	
	10%	DB	0.5	0.8	3.1	7.9	15.3	21.2	25.1	24.0	19.5	13.1	8.2	3.1	(25)	
		MCWB	-0.2	-0.2	1.3	4.6	10.1	14.6	18.0	17.5	15.7	10.7	7.1	2.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	7.3	7.2	11.6	16.5	20.7	22.6	22.0	20.7	17.1	14.2	10.1	(27)	
		MCDB	9.8	8.0	8.9	13.6	20.7	25.3	27.9	26.9	23.6	18.9	15.1	10.7	(28)	
	2%	WB	5.0	4.4	4.8	9.0	14.1	18.5	21.4	20.6	19.1	14.9	12.1	6.9	(29)	
		MCDB	5.6	5.0	6.2	11.4	18.0	22.6	26.2	25.1	21.8	16.9	12.9	7.5	(30)	
	5%	WB	2.1	1.7	3.0	6.8	12.3	16.9	20.2	19.6	17.6	12.9	10.0	4.4	(31)	
		MCDB	2.7	2.5	4.6	9.2	17.0	21.5	24.2	23.5	20.3	14.7	11.0	4.8	(32)	
	10%	WB	-0.1	0.0	1.5	5.0	10.6	15.3	19.0	18.5	16.2	10.9	7.1	2.5	(33)	
		MCDB	0.4	0.7	2.8	7.5	14.8	20.0	23.2	22.3	19.0	12.9	8.0	3.1	(34)	
(35) Mean Daily Temperature Range	MDBR		11.1	12.5	11.8	10.2	12.6	13.1	13.2	13.2	12.9	10.4	8.4	8.3	(35)	
	5% DB	MCDBR	11.0	11.7	13.4	14.0	19.9	19.4	17.3	16.9	15.7	13.9	10.7	9.2	(36)	
		MCWBR	10.9	11.0	10.8	9.9	12.1	10.8	9.3	9.5	10.4	10.7	10.1	8.9	(37)	
	5% WB	MCDBR	10.7	11.0	11.7	12.2	16.1	14.5	13.1	12.7	12.8	12.3	10.5	9.0	(38)	
		MCWBR	10.7	10.6	10.2	9.5	10.8	8.9	8.0	8.1	9.4	10.2	10.1	9.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.308	0.340	0.367	0.365	0.383	0.411	0.393	0.363	0.332	0.309	0.300	(40)	
	taud		2.274	2.293	2.285	2.324	2.405	2.373	2.328	2.366	2.453	2.507	2.479	2.320	(41)	
	Ebn at Noon		746	838	868	880	895	878	845	845	839	810	748	694	(42)	
	Edh at Noon		73	91	109	117	112	117	121	111	92	74	61	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.90	1.64	2.82	3.95	4.75	5.05	5.13	4.27	3.25	1.99	1.05	0.66	(44)	
	RadStd		0.09	0.22	0.26	0.45	0.35	0.56	0.40	0.30	0.24	0.18	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	+1.22	N/A	+0.39	+0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page